



ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

MECHANICAL TECHNOLOGY

MARKING GUIDELINE

MARKS: 100

Marking guideline for the Academic Central Grade 11 Mechanical Technology practice examination.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. It is not a Department of Basic Education marking guideline.

How to mark: each ✓ is one mark. Penalise once per question for missing units. Accept other correct answers and small rounding differences.

QUESTION 1

1 1.1 B ✓✓ 1.2 A ✓✓ 1.3 A ✓✓ 1.4 A ✓✓ 1.5 A ✓✓ [10]

QUESTION 2

- 2.1** Any FOUR: wear safety glasses ✓; remove the chuck key before starting ✓; no loose clothing or jewellery, tie back long hair ✓; stop the machine before measuring or cleaning ✓; remove swarf with a brush, never by hand; know where the emergency stop is (4)
- 2.2** (a) Tightens nuts and bolts to the manufacturer's specified torque ✓ (3)
 (b) Finds where compression leaks out of a cylinder (valves, piston rings or head gasket) ✓
 (c) Measures the compression pressure in each cylinder to find worn rings or leaking valves ✓
- 2.3** $12 + 0,5 + 0,27 \checkmark = 12,77 \text{ mm } \checkmark$ (2)
- 2.4** Any FOUR: test for leaks with soapy water ✓; fit flashback arrestors ✓; never use oil or grease on oxygen fittings ✓; wear goggles with the correct shade of filter lens ✓; open cylinder valves slowly; follow the correct lighting and shut-down order; keep cylinders chained upright (4)
- 2.5** Acetylene is dissolved in acetone; if the cylinder lies down, acetone can flow out and cause a dangerous flame ✓. An upright, chained cylinder cannot fall and break its valve ✓. (2)

[15]

QUESTION 3

- 3.1** (a) Hardening: heat to about 30–50 °C above the upper critical temperature, soak, and quench quickly in water or oil ✓; makes steel hard and wear resistant (but brittle) ✓. (8)
 (b) Tempering: reheat hardened steel to below the lower critical temperature (about 200–600 °C) and cool ✓; reduces brittleness and increases toughness ✓.
 (c) Annealing: heat above the upper critical temperature, soak, and cool very slowly in the furnace ✓; makes steel as soft as possible, relieves stress and improves machinability ✓.
 (d) Normalising: heat above the upper critical temperature, soak, and cool in still air ✓; refines the grain structure and relieves internal stresses ✓.
- 3.2** Giving low-carbon steel a hard, wear-resistant outer skin ✓ while the core stays soft and tough ✓. Any ONE: gears, camshafts, gudgeon pins, crankshaft journals ✓ (3)

- 3.3** (a) A file slides over hard steel without cutting ✓ but bites easily into soft metal, so it shows how hard the metal is ✓. (4)
 (b) Steel rings clearly when struck with a hammer ✓, while cast iron gives a dull sound ✓.

[15]

QUESTION 4

- 4.1** $A = \pi \times 30^2 \div 4 = 706,86 \text{ mm}^2$ ✓ (7)
 (a) $\sigma = F \div A = 80\,000 \div 706,86 \text{ ✓} = 113,18 \text{ MPa ✓}$
 (b) $\varepsilon = \sigma \div E = 113,18 \div 210\,000 \text{ ✓} = 5,39 \times 10^{-4} \text{ ✓}$
 (c) $\Delta L = \varepsilon \times L = 5,39 \times 10^{-4} \times 1\,500 \text{ ✓} = 0,81 \text{ mm shorter ✓}$
- 4.2** $UDL = 5 \times 8 = 40 \text{ kN acting at 4 m from A ✓}$ (7)
 Moments about A: $\Sigma CWM = \Sigma ACWM \text{ ✓}$
 $R_B \times 8 = (30 \times 2) + (50 \times 5) + (40 \times 4) \text{ ✓✓} = 470 \text{ kN}\cdot\text{m}$
 $R_B = 58,75 \text{ kN ✓}$
 $R_A = (30 + 50 + 40) - 58,75 \text{ ✓} = 61,25 \text{ kN ✓}$
- 4.3** $\Sigma F_x = 120 \cos 30^\circ + 80 \cos 135^\circ + 50 \cos 270^\circ \text{ ✓} = 103,92 - 56,57 + 0 =$ (8)
 $47,35 \text{ N ✓}$
 $\Sigma F_y = 120 \sin 30^\circ + 80 \sin 135^\circ + 50 \sin 270^\circ \text{ ✓} = 60 + 56,57 - 50 = 66,57$
 N ✓
 $R = \sqrt{(47,35^2 + 66,57^2)} \text{ ✓} = 81,69 \text{ N ✓}$
 $\theta = \tan^{-1}(66,57 \div 47,35) \text{ ✓} = 54,58^\circ \text{ anticlockwise from the positive x-axis}$
 (first quadrant) ✓
- 4.4** The greatest stress a material can carry ✓ and still return to its original length when the load is removed ✓. Beyond it, the material is permanently deformed ✓. (3)

[25]

QUESTION 5

- 5.1** (a) $N_C = N_A \times T_A \div T_C \text{ ✓} = 1\,200 \times 24 \div 72 \text{ ✓} = 400 \text{ r/min ✓}$ (6)
 (b) $VR = T_C \div T_A = 72 \div 24 = 3 : 1 \text{ ✓}$
 (c) It makes A and C turn in the same direction ✓ without changing the velocity ratio ✓
- 5.2** (a) $N_2 = N_1 \times D_1 \div D_2 \text{ ✓} = 1\,440 \times 150 \div 400 \text{ ✓} = 540 \text{ r/min ✓}$ (5)
 (b) $v = \pi DN \div 60 = \pi \times 0,15 \times 1\,440 \div 60 \text{ ✓} = 11,31 \text{ m/s ✓}$

- 5.3 (a) $A_{\text{small}} = \pi \times 0,025^2 \div 4 = 4,91 \times 10^{-4} \text{ m}^2 \checkmark$ (6)
 $p = F \div A = 300 \div 4,91 \times 10^{-4} \checkmark = 611,15 \text{ kPa} \checkmark$
 (b) $A_{\text{large}} = \pi \times 0,2^2 \div 4 = 0,0314 \text{ m}^2 \checkmark$
 $F = p \times A = 611155 \times 0,03142 \checkmark = 19200 \text{ N (19,2 kN)} \checkmark$
- 5.4 Any THREE: the wedge action grips better, so there is less slip $\checkmark$; it transmits more power $\checkmark$; it runs more quietly $\checkmark$; it works on short centre distances; it is less likely to come off the pulley (3)

[20]

QUESTION 6

- 6.1 Preventive maintenance is planned, regular servicing $\checkmark$ done before a breakdown happens, e.g. oil changes $\checkmark$. Corrective maintenance is repairing or replacing parts $\checkmark$ after a fault or breakdown has happened $\checkmark$. (4)
- 6.2 Any THREE, one mark for the cause and one for prevention: (6)
 Porosity: gas trapped from dirty metal or damp electrodes $\checkmark$ – clean the metal, use dry electrodes $\checkmark$
 Undercut: current too high or wrong electrode angle $\checkmark$ – lower the current, use the correct angle $\checkmark$
 Slag inclusion: slag not removed between runs $\checkmark$ – chip and brush every run $\checkmark$
 Cracks: cooling too fast – preheat and cool slowly
 Incomplete penetration: current too low or travel too fast – increase current, weld slower
- 6.3 Any FIVE: spring washer $\checkmark$; lock nut $\checkmark$; castle nut and split pin $\checkmark$; tab washer $\checkmark$; nylon-insert (Nyloc) nut $\checkmark$; thread-locking compound; locking wire (5)

[15]

TOTAL: 100

END OF MARKING GUIDELINE